MILITARY SPECIFICATION SHEET

ELECTRON TUBE, VOLTAGE REGULATOR

TYPE 6213A

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.

DESCRIPTION: Glow discharge, subminiature.

Outline:	See figure 1.
Base:	3-lead flat press.
Envelope:	T-3.
Cathode:	Glow discharge.
Base Connections:	Pin No: 1 3 5
	Element: k a k

ABSOLUTE MAXIMUM RATINGS:

Parameter:	Ebb	Ionization Voltage (Total Darkness)	Ionization Voltage (Amb. Light)	Operating Voltage	Operating Current	T. Amb.	Alt.
Unit:	Vdc	Vdc	Vdc	Vdc	mAdc	°C	Ft.
Maximum:	---	---	---	133	2.5	+90	60,000
Minimum:	---	200	180	127	1.0	-55	----

TEST CONDITIONS: (Note 1)

GENERAL:

Qualification - Not Required

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:03Z
Check the source to verify that this is the current version before use.

Method	Requirement or test	Conditions	AQL (percent defective)	Insp. level or code	Symbol	Limits		Units
						Min	Max	
	Qualification							
1031	Variable frequency Vibration (1)	Rp = 10,000 ohms Ebb/Ib = 2 mAdc	---	---	---	---	50	mVac
	Quality conformance inspection, part 1 (Note 2)							
3347	Ionization voltage (1)	Ebb/Ib = 1 to 2.5 mA; RL = 30,000 ohms; Illumination = 5 - 50 ft. candles	0.65	II	Ez(1)	160	180	Vdc
3337	Voltage drop (1)	Ebb/Ib = 1.0 mA; RL = 30,000 ohms	0.65	II	Etd(1)	127	133	Vdc
3337	Voltage drop (2)	Ebb/Ib = 2.5 mA; RL = 30,000 ohms	0.65	II	Etd(2)	127	133	Vdc
3335	Regulation	Etd(1)-Etd(2)	0.65	II	Reg.	---	±1.0	Vdc
3278	Noise	Ebb/Ib = 2.5 mA; RL = 5000 ohms	0.65	II	Eb	---	20	mVac
1201	Shorts & discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection part 2, Note 2							
3347	Ionization voltage (2)	Ebb/Ib = 1 to 2 mA; RL = 30,000 ohms; total darkness	---	---	Ez(2)	---	200	Vdc
---	Ionization time	Ebb = 200 Vdc; RL = 30,000 ohms; total darkness note 3	---	---	ti	---	1.0	Sec.
---	Operation test	Ebb = 180 Vdc; RL = 30,000 ohms; note 4	1.5	Code G	---	---	---	---
3305	Leakage current	Ebb = 50 Vdc; RL = 3,000 ohms	---	---	Lib	---	20	μAdc
1031	Low frequency vibration	Ebb/Ib - 2 mAdc; RL = 10,000 ohms; F = 40 Hz; G = 15	---	---	Ep	---	50	mVac

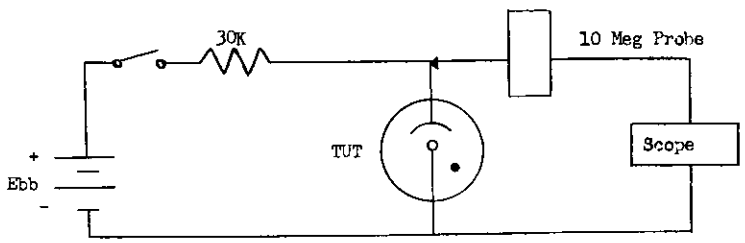
Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:03Z
Check the source to verify that this is the current version before use.

Method	Requirement or test	Conditions	AQL (percent defective)	Insp. level or code	Symbol	Limits		Units
						Min	Max	
	Quality conformance inspection, part 2, Note 2 (continued)							
1116	Lead fatigue		4.0	S3	---	4.0	---	Arcs
2126	Glass strain		2.5	I	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
1501	Intermittent life test	Ebb/Ib = 2.5 mAdc; Group A;t = 500 hours (min)						
---	Intermittent life test end points (500 hrs.)	note 2						
	Ionization voltage (1)				Ez(1)	---	180	Vdc
	Voltage drop (1)				Etd(1)	127	133	Vdc
	Voltage drop (2)				Etd(2)	127	133	Vdc
	Regulation				Reg.	---	±1.5	Vdc

NOTES:

1. A fixed resistor may be used and Ebb varied to give desired current.
2. Ionization between leads at base of tube, within envelope, at any time during operation on any of the tests shall be cause for rejection.
3. Ionization time is defined as the length of time from initial application of voltage to the tube until the tube starts to conduct. The tubes shall be tested in the following circuit:

Ionization Time Test Circuit



Upon closing of the switch an Ebb of specified voltage from a stabilized power supply shall suddenly be applied to the tube. The length of time from the initial application of voltage until the tube conducts shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:03Z
Check the source to verify that this is the current version before use.

be observed on a suitable oscilloscope (such as Tektronix 531 or equal) or measured with a suitable timing device. A tube with an ionization time greater than limit shown shall be considered a reject. Prior to test, the tubes shall be in total darkness for a minimum period of 24 hours without any voltages applied.

4. Subject each tube to 20 ON-OFF operations by suddenly applying Ebb. During each ON cycle, the tube voltage drop shall be observed and shall be within the limits for tube voltage drop (2).

User activities:
Navy - AS, OS, MC, CG, SH

Preparing activity:
Navy - EC

(Project 5960-N888)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:03Z
Check the source to verify that this is the current version before use.

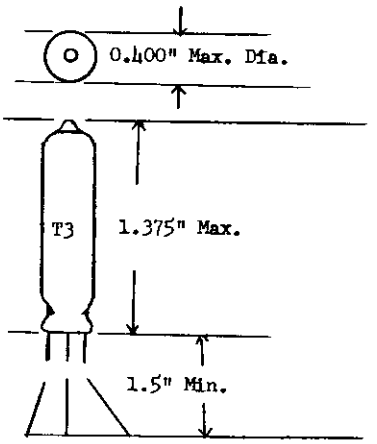


FIGURE 1. Outline for Type 6213A

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:03Z
Check the source to verify that this is the current version before use.